Apex Institute of Technology

Computer Science & Engineering

Experiment 5

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Branch: B.E. CSE (AIML) **Section: 24AIT_KRG-G1**

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Subject Name: Database Management System **Subject Code: 24CSH-298**

AIM: This experiment focuses on the use of conditional logic in SQL using the MOD (modulus) operator to determine whether employee salaries are odd or even. The SQL queries demonstrate how numerical data can be analyzed and categorized using conditional logic in a database.

OBJECTIVES:

- To understand the concept of conditional logic in SQL.
- To use the MOD operator to determine odd and even numbers.
- To write SQL queries that classify employee salaries as odd or even.
- To analyze numerical data stored in a database table.

SOFTWARE REQUIREMENTS:

- Oracle FreeSQL / Oracle Database Express Edition (Oracle XE)
- Oracle SQL Developer

PROCEDURE/STEPS:

1. Start the system and log in.
2. Open Oracle Free SQL / Oracle Live SQL.
3. Enable output using SET SERVEROUTPUT ON.
4. Write the PL/SQL program in the worksheet.

5. Execute the program.
6. Observe the output.
7. Save the work and take screenshots.

PRACTICAL/EXPERIMENTAL STEPS:

```
DROP TABLE IF EXISTS employees;

CREATE TABLE employees (
   emp_id NUMBER PRIMARY KEY,
   emp_name VARCHAR2(50),
   salary NUMBER
);

INSERT INTO employees VALUES (1, 'Jaskaran', 25000);
INSERT INTO employees VALUES (2, 'Yuvi', 30001);
INSERT INTO employees VALUES (3, 'Anhad', 45000);
INSERT INTO employees VALUES (4, 'Manan', 52003);
INSERT INTO employees VALUES (5, 'Kartik', 61000);

COMMIT;

SELECT emp_id, emp_name, salary
FROM employees
WHERE MOD(salary,2) = 0;

SELECT emp_id, emp_name, salary
FROM employees
WHERE MOD(salary,2) <> 0;

SELECT emp_id, emp_name, salary,
CASE
   WHEN MOD(salary,2) = 0 THEN 'Even Salary'
   ELSE 'Odd Salary'
END AS salary_type
FROM employees;
```

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OUTPUT:

[SQL Worksheet]*

```sql
DROP TABLE IF EXISTS employees;
CREATE TABLE employees (
    emp_id NUMBER PRIMARY KEY,
    emp_name VARCHAR2(50),
    salary NUMBER
);

INSERT INTO employees VALUES (1, 'Jaskaran', 25000);
INSERT INTO employees VALUES (2, 'Yuvi', 30001);
INSERT INTO employees VALUES (3, 'Anhad', 45000);
INSERT INTO employees VALUES (4, 'Manan', 52003);
INSERT INTO employees VALUES (5, 'Kartik', 61000);

COMMIT;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
SQL> SELECT emp_id, emp_name, salary,
     CASE
         WHEN MOD(salary,2) = 0 THEN 'Even Salary'
         ELSE 'Odd Salary'...
Show more...

EMP_ID EMP_NAME   SALARY SALARY_TYPE
------ ---------- ------ -------------
1      Jaskaran   25000  Even Salary
2      Yuvi       30001  Odd Salary
3      Anhad      45000  Even Salary
4      Manan      52003  Odd Salary
5      Kartik     61000  Even Salary

Elapsed: 00:00:00.005
5 rows selected.
```

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[SQL Worksheet]*

```sql
DROP TABLE IF EXISTS employees;
CREATE TABLE employees (
    emp_id NUMBER PRIMARY KEY,
    emp_name VARCHAR2(50),
    salary NUMBER
);

INSERT INTO employees VALUES (1, 'Jaskaran', 25000);
INSERT INTO employees VALUES (2, 'Yuvi', 30001);
INSERT INTO employees VALUES (3, 'Anhad', 45000);
INSERT INTO employees VALUES (4, 'Manan', 52003);
INSERT INTO employees VALUES (5, 'Kartik', 61000);

COMMIT;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download | Execution time: 0.008 seconds

	EMP_ID	EMP_NAME	SALARY
1	1	Jaskaran	25000
2	3	Anhad	45000
3	5	Kartik	61000

I/O ANALYSIS:

Input:

- Employee ID
- Employee Name
- Salary

Output:

- Employees with even salaries
- Employees with odd salaries
- Salary classification (Even Salary / Odd Salary)

LEARNING OUTCOMES:

- Understood how SQL conditional logic works.
- Learned to use the MOD function for identifying odd and even numbers.
- Practiced filtering and categorizing data using SQL queries.
- Improved understanding of SQL query execution in Oracle SQL Developer.